

VestiCare: A Holistic and Smart Digital Platform to Improve the Vestibular Rehabilitation Experience

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ABSTRACT

The goal of SDG 3 (Good Health and Well-being) is to promote healthy lives for all, at all ages. An often-neglected health concern, vestibular disorders, can negatively affect people's balance and spatial stability. It requires consistent and accurate rehabilitation exercises for treatment. However, the low accessibility and the lack of motivation among patients make vestibular rehabilitation less effective. To tackle these challenges, we first conducted literature reviews and interviews with specialists to discover the user needs. Then, we defined user personas and journeys to develop both low- and high-fidelity prototypes. Our final solution is a digital platform comprising a mobile app and web portal for patients and healthcare providers (HCPs). We utilised phone sensors and AI to increase vestibular rehabilitation accessibility, gamified exercises to boost patient motivation, and incorporated an efficient communication channel for better HCP-patient communication. In general, usability tests demonstrated that our platform was user-friendly, accessible, and motivating.

CCS CONCEPTS

• **Human-centered computing**; • **Human-computer interaction (HCI)**; **Interaction design**;

KEYWORDS

Vestibular rehabilitation, vestibular disorders, vertigo, chronic dizziness, digital healthcare, gamification, mobile application, web portal, AI

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1 INTRODUCTION

The vestibular system is a sensory system that helps regulate our sense of balance and orientation in space. Its receptors lie in the inner ear next to the auditory cochlea [1]. Hypofunction in this area can lead to symptoms such as vertigo, nausea, chronic dizziness, postural instability, and even hearing loss [2]. In fact, research suggests that patients with vestibular disorders are 12 times more likely to fall than people without the disorders [3]. Vestibular disorders can occur in people across all age groups, and it is estimated that between 53 and 95 million adults across Europe and the US have peripheral (inner ear receptors) vestibular disorders [4]. There is also a significant increase with age, with about 50% of those 60 years and above suffering from such disorders [3].

Vestibular rehabilitation (VR) is an exercise-based therapy that involves a series of physiological exercises, such as gaze stabilisation, head movement and balance training that can help patients become habituated to movements that trigger dizziness and other symptoms. According to the Clinical Practice Guidelines for vestibular hypofunction, patients are recommended to perform exercises consistently and accurately for 15 to 20 minutes every day [5], and the whole process of VR usually lasts from weeks to months.

So far, there are two well-documented challenges identified during the rehabilitation process: *low accessibility* for conducting proper VR and *low motivation* for patients to engage in exercises long-term. According to a European survey across 22 countries, nearly half of the therapists reported that VR was difficult to access [6] because traditional VR can bring physical and financial difficulties for patients to attend clinics regularly [7]. In terms of motivation, patients may easily get fatigued or bored, as VR is a long-term process and exercises can be repetitive. A study on chronic dizziness reported that long-term adherence to exercise programmes was below 50% [8]. Thus, there is a significant demand for a comprehensive solution that addresses the problems.

The present design aims to bridge the gaps of low accessibility and motivation in VR process with the help of digital technology. To achieve this, we conducted literature reviews and semi-structured interviews to understand key pain points. This built the foundation for our design process, which includes defining user stories, creating personas, wireframing, as well as developing low- and high-fidelity

prototypes. We also conducted usability tests with specialists to attain feedback on our design. While this disorder is invisible, its implications are disabling. Thus, we believe our designs can contribute to the UN's sustainable goal of Good Healthcare and Well-being (goal number 3).

2 LITERATURE REVIEW

It is suggested that digital technology may help increase the accessibility of VR by changing the traditional in-person process to a remote one. A study found that more than 80% of therapists agreed that digital health was an effective treatment delivery method because it can provide patients with more personalised treatment plans and also reduce economic costs [9]. At present, some attempts have been made to use mobile-based digital technology for VR. For instance, Meldrum et al. developed a mobile app specifically for VR, and a head sensor to detect head motions to provide in-time feedback [10]. They achieved high usability ratings and their participants had better symptom improvement, compared to those who engaged in non-digital VR. However, we argued that purchasing and connecting external devices can cause extra financial and technical difficulties for patients, which can make VR less accessible. The camera and sensors embedded in mobile phones, such as accelerometers and gyroscopes, are already able to detect eye gaze [11] and orientation [12]. Thus, we believe an application on the phone is sufficient to increase the accessibility of VR, hence, a head sensor may not be necessary for treatment.

Gamifying the rehabilitation process has been applied to increase patient motivation. For instance, D'Silva et al. developed a gamified VR platform with customisable visual stimuli based on patients' severity to increase treatment adherence [13]. Additionally, Bonnechere et al. showed that patients' rehabilitation treatment through commercial video games was as effective as administering traditional treatment [14]. However, their games were designed for horizontal displays (e.g., desktops), which may not be as accessible for patients to use regularly in the long run. As such, it is necessary to develop novel gamified exercises tailored for mobile settings.

Existing VR platforms have already incorporated important elements of the VR process, including remote at-home exercises, tailored training plans and self-tracking mechanisms. However, thus far, studies have not involved healthcare providers (HCPs) in the platform, meaning that patients' platform is apart from HCP's supervision; hence, patients may lack proper guidance and monitoring during the remote training. We argue that HCP involvement within a digital platform can be important in remote VR process due to the following reasons: First, involving the HCP will increase patients' trust towards using the platform, because the treatment process will be reviewed by HCPs. Additionally, with synchronised monitoring data shown on the provider's interface, they can offer immediate adjustment of treatment if patients do not respond well to the exercises [15]. More importantly, providing remote support can eliminate the need for in-person consultations, which helps save time for both parties [16]. Reducing travel could also benefit patients whose symptoms can be triggered by commuting. Thus, along with patients, we considered the HCPs to be another main user group of the VR platform.

3 USER RESEARCH AND REQUIREMENTS

This study has been approved by the UCLIC Research Ethics Committee with the project ID number: UCLIC_2021_015_Marquardt_IxD. The approval did not include permission to directly contact patients. Thus, we conducted semi-structured interviews with 4 vestibular specialists. They came from worldwide, with a mean working experience of more than 10 years. Interviews were conducted online via Microsoft Teams. We began our interviews by exploring the challenges of VR process, followed by discussing their use of digital healthcare platforms, and finished with the evaluation of HCP-patient communication.

A wealth of qualitative data was analysed via bottom-up thematic analysis with the help of an affinity diagram. We summarised the following key requirements to be considered in our design.

It would be important to create a platform that can be easily accessed and time-efficient for both patients and HCPs. Patients' needs, such as customised exercises, clear instructions, and continued HCP support were highlighted, while for HCPs, it was important to have a simple and efficient interface for assigning exercises and managing patient data. We also found it is important to reduce the potential visual impact on patients by implementing pale background colours and interfaces with mild transitions, given that motions and high-contrast colours may trigger symptoms. To motivate patients, many interviewees mentioned that in addition to creating engaging exercises, it is important to show patients their improvement and progress via data visualisations. Lastly, the communication between HCPs and patients is very important for successful VR, with all interviewees rating the importance above 8 (out of 10). However, it was noted that instant messaging is not an ideal way of HCP-patient communication, as this may result in HCPs losing track of patients' messages, which can be overwhelming for HCPs. Thus, we deduced that it was important to have a solution which can increase the efficiency of HCP-patient communication, without increasing treatment burdens for both parties.

4 DESIGN PROCESS

Feedback from interviews helped us create a storyboard to illustrate users' stories and summarise key pain points. As shown in Figure 1, user personas were then created to define two main user groups: vestibular disorder patients and HCPs who treat them. We also created a user journey using the digital platform and wireframes to demonstrate user flows. The final step before prototyping was defining our targeted scenarios. Firstly, our platform is not designed to be freely downloaded by the public, access to this app would be protected with a specialised code that would be with the HCP. Patients would only get access once their provider deemed them eligible for utilising the platform. Secondly, our platform is created specifically for rehabilitation, which means that it is not a medium for diagnosing patients.

4.1 Low-Fidelity Prototype

Based on the research and user requirements, we created a cutting-edge and user-centred platform, which consists of a mobile app and a web portal for patients and HCPs respectively, due to the two



Figure 1: Personas of the two user groups.

groups' different requirements. Detailed functionalities are introduced in the following paragraphs, with each paragraph addressing one main goal.

4.1.1 Increasing Accessibility Via Digitalisation and Incorporating Phone Camera and AI. We aim to create accessible platforms for both user groups. Given that specialists mentioned the need for personalised and accessibility, we created an application where patients can check rehabilitation plans and perform exercises remotely and flexibly. For HCPs, a web-based portal was created, as most interviewees mentioned they use desktops in a working environment. HCPs can use the platform to manage patients by adding new patients and accessing patients' progress and data. A crucial feature of this system is that patients' rehabilitation plan is tailored to their conditions. After the first clinical visit, the HCP can upload patients' diagnoses onto the web portal, and embedded specialised AI will automatically generate the patients' rehabilitation plan (see Figure 3, interface 8). For this feature, specialists mentioned that they still wish to have some levels of control, and so HCPs can adjust the generated plan to ensure the correct exercises and dosing. Finally, HCPs can assign the plan, and patients would get notified on the app. These processes explain the integration of two interfaces (see Figure 2 for detailed interaction), emphasising the importance of designing these platforms in conjunction. Lastly, as discussed in the literature review, the use of external devices can be

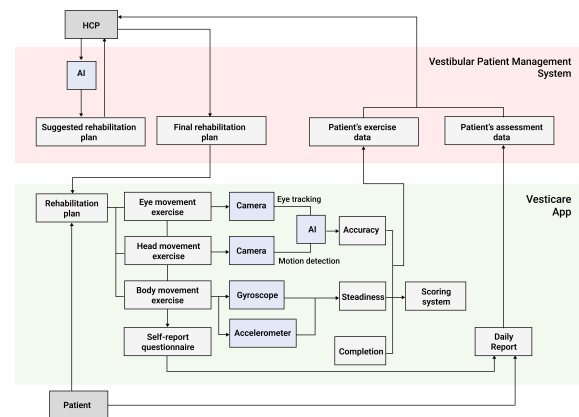


Figure 2: Diagram illustrating the interaction of two interfaces, the Vestibular Patient Management System (i.e., the HCP web portal) and VestiCare APP.

replaced by embedded functions in the phone. Thus, to visualise interview feedback regarding exercise monitoring, we applied phone cameras to detect eye movement to ensure patients are performing the exercises accurately. For instance, it can detect whether patients' eyes or heads move correctly. If patients are not performing the exercises correctly (e.g., not in the camera, being distracted), a message will pop out to alert them. An accuracy rate will then be generated based on patients' performance.

4.1.2 Motivating Patients Via Tree Planting Game. To address specialists' feedback regarding low patient motivation, we created gamified versions of existing vestibular exercises. The idea of the tree planting game was implemented as we observed that the plantation process followed similar patterns to VR. In both scenarios, users must be patient and consistent throughout the process to ensure a successful outcome is achieved. We used three types of exercises as examples for our gamification concept. We first gamified eye movement training, where patients need to water the sapling by following the movement of the sprinkling pot with their eyes (see Figure 3, interface 2). After a tree grows and several apples appear, patients go to the head movement training. Here, patients need to make apples fall from the tree by moving their head up and down. The third example is head rotation training, where patients should wash the apples by rotating their head in the direction of the sprinkling pot. A unique feature of our gameplay is that it connects different exercises together, which not only offers a continuous rehabilitation experience but also redefines VR as an enjoyable experience rather than a monotonous task. Given the emphasis on tracking symptom improvements during interviews, patients are asked to complete a self-reported questionnaire (covering dizziness, nausea, imbalance, and anxiety) after the exercises to rate how they feel on a scale of 1 to 10. The data will be stored to generate a daily report on the patient's Profile page, so they can see improvements and gain a sense of accomplishment.

4.1.3 Effective HCP-Patient Communication Via Appointment Booking and AI Support. A key point mentioned during interviews is the

effective communication between HCPs and patients. Hence, we created an appointment feature (see Figure 3, interface 4), which allows patients to view the HCP's availability and book online or in-person sessions, without increasing HCPs' workload. Importantly, patients can leave a note for the HCP by describing their symptoms via the speech-to-text function or text typing while booking the appointment. This pre-consultation information can help providers prepare for the session and improve session efficiency. Additionally, we added an AI chatbot so that patients can go over quick questions without waiting. Note that the answer from AI will be strictly monitored to ensure its correctness, and the questions which cannot be answered by the AI chatbot will be referred to the HCP.

4.2 Usability Testing

Again, due to the ethical approval constraint, we could not invite patients to the usability testing. We then conducted usability testing for the prototypes with vestibular specialists ($N = 8$) worldwide because we believe specialists not only understand the patients' situation but can also be the target user of the HCP web interface. Three of them were recruited from previous interviews and five were novel to our design idea.

The usability testing tasks were designed to assess whether participants could navigate through the app successfully and independently. The process was divided into two sections. Firstly, participants performed specific tasks on the patient's interface (i.e. the mobile application). Tasks included navigating to the rehabilitation plan, engaging with the gamified exercises, completing the self-reported questionnaire, booking an appointment with a pretend HCP and using the AI chatbot to find quick information. We also asked participants to think aloud as they interacted with different elements. In the next section, participants interacted with the web interface, where they were asked to add new exercises for a patient, access the appointment schedule, review notes left by patients, and evaluate patients' self-reported data.

4.2.1 Specialists' Feedback. In terms of constructive feedback, specialists mentioned that adding exercises which support body movement will help increase accessibility, because many patients experience dizziness while performing daily activities, such as bending over to pick up utensils from a dishwasher and placing them overhead. This led us to apply sensors in the phone to support such exercises (see 4.3.1). They also mentioned that patients may lack awareness of their symptoms and situations, resulting in frustration and anxiety. Thus, accessing awareness-focused content was crucial, leading us to create a learning hub (see 4.3.1). Interestingly, the community feature received the most criticism. This feature was designed because this might help patients connect with like-minded people so as to support and encourage each other. However, specialists pointed out that there was a high chance of it turning negative. For example, the community might become a place to complain. This feedback aligns with research discovered that the information sought by cancer patients can be distorted, which can negatively impact patients and their perceptions [18]. Thus, we decided to remove the community feature.

The consensus was that the rehabilitation plan user flow was intuitive and easy to follow. They found the theme colours soothing as well as dizziness- and migraine-friendly. All specialists agreed

that adding elements of gamification to the exercises can increase motivation by creating a sense of fun, especially for younger patients. When evaluating the appointment feature, specialists liked that patients could leave a note before the appointment because it helps them prepare for the session, which can improve work efficiency. Moreover, all specialists agreed that having an AI chatbot to answer quick questions about patients' concerns is a valuable feature to have, as long as the answers are appropriate.

For HCP's interface, all specialists agreed that a web display is a suitable mode for HCP. They liked the fact that the HCP interface is simple and easy to follow. The features of adding exercises and monitoring patient data are straightforward and accessible.

To quantify the evaluation of our design, we asked specialists to rate from 0 to 10 on each feature and created an evaluation matrix (see Table 1). Our design scored highest on overall user-friendliness, followed by accessibility and motivation. The scores illustrated that our design satisfied the goals of resolving the issues of low accessibility and the lack of motivation. As discussed above, communities received the lowest score given the challenges of implementing it. Finally, we applied the Net Promoter Score (NPS) to evaluate specialists' enthusiasm and motivation to use our design [19]. The high NPS reflected that most specialists were open to recommending our platform to patients and other professionals.

4.3 High-Fidelity Prototype

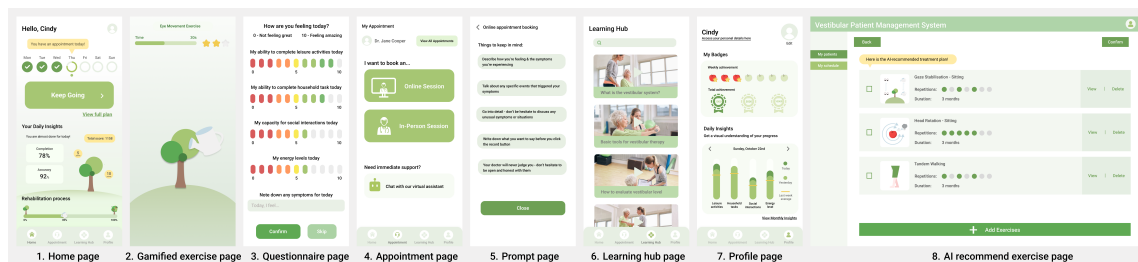
Based on usability tests, we implemented participants' feedback and polished our existing designs to a high-fidelity prototype. The following paragraphs indicate the changes made after the usability testing.

4.3.1 Further Increasing Accessibility by Applying Phone Sensors and Adding Educational Content. We implemented the phone's accelerometer and gyroscope to assess the phone's orientation and stability. This is to support body movement exercise (see section 4.2.1 for explanations). Specialists mentioned that some patients need to perform an exercise by steadily lifting an object from the floor over their head. Thus, while holding the phone during this exercise, the accelerator and gyroscope sensors in the phone can detect whether patients' movements are smooth, and the lifting height is reached. Additionally, our usability tests revealed the critical need for educating patients on their specific vestibular concerns. Thus, to make VR information further accessible, we incorporated a learning hub section, which replaced the community (see 4.2.1 for explanations), where specialists can post evidence-based information regarding vestibular disorders to help patients deal with the uncertainty of living with this disorder (see Figure 3, interface 6).

4.3.2 Strengthening the Rewards to Motivate Patients. Based on feedback related to encouraging patients, we created a scoring system to increase the patient's motivation to complete rehabilitation. During the exercise, we added a feature wherein patients would receive a star for every ten seconds of persistent training. The higher the completion rate and accuracy of training, the more scores they earn. As patients make progress, they can click the yellow bubble icons on the Home Page to help their tree grow (see Figure 3, interface 1). On the Profile page, badges can be earned by completing daily exercises and fulfilling certain scores. We also

Table 1: Evaluations of Each Feature and NPS

	Overall User-friendliness	Accessibility of both interfaces	Tools for Motivation	HCP-patient communication	Community	NPS
Mean Rating	9.75	8.5	8.75	7.5	6	8.5

**Figure 3: Interfaces of the VestiCare APP and Vestibular Patient Management System.**

added weekly and monthly reports of patients' progress (see Figure 3, interface 7). Finally, from feedback gathered within usability tests, we redesigned the content of the post-exercise self-report questionnaire from the severity of four symptoms to the rating of how capable they can perform daily activities [17], as reflecting on symptoms may be frustrating for some patients. The revised questionnaire is more encouraging and generalisable (see Figure 3, interface 3).

4.3.3 Optimising the Pre-consultation Note-Leaving. We added a prompt list (see Figure 3, interface 5) as a guide for patients so that they can efficiently describe their concerns while leaving a note for the HCP before the appointment.

5 CONCLUSION AND FUTURE DIRECTIONS

Disorders in the vestibular system can have debilitating consequences for patients. Given the low accessibility of traditional VR and the lack of motivation to perform exercises, our project bridged the gaps by introducing digitalised and technology-focused design solutions for both patients and HCPs. Our design stands out by a) incorporating phone sensors and specialised AI to increase VR accessibility, b) redefining the rehabilitation experience using gamified elements to motivate patients, and c) streamlining the communication between HCPs and patients in remote VR process.

While all specialists appreciated our design, there are some limitations that can be tackled with future research. In terms of accessibility, given that a great population of this disorder are the elderly [3], there is a concern regarding the usability of our platform within the older population. Thus, in the future, a more comprehensive plan could be deployed that involves features and resources that are user-group-specific. Another concern was related to the self-reported questionnaire. While improvement awareness is important, some patients may feel hyper-aware of their performance, which can make them feel worse if they do not notice improvement. Future iterations can account for other mediums of progress and improvement tracking to ensure patients do not feel worse about their condition. Finally, given the close link between anxiety and

chronic dizziness [20], our future direction is to incorporate mental health support to help patients acknowledge their emotions and build acceptance towards their discomfort and distress.

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